



S1Gate

The Universal System-1 Pre-Commit Decision Layer & Remote MCP Server
for AI Coding Harnesses

⚡ Sub-Second Latency (~1.2s)

📊 100% CVE Catch Rate (12/12)

🛡️ IBM Bob Actor-Critic Remediation

🌐 Hosted Remote SSE MCP

Heavy LLM Reasoning Destroys Commit Flow

Modern AI agents (IBM Bob, Claude Code) are built for complex multi-file architecture synthesis (System-2). Applying them blindly to micro-commits creates catastrophic friction.

WITHOUT S1GATE

FLAWED PATTERN

15 - 30s

Autoregressive Latency per Micro-Commit

- ❑ **Developers bypass hooks:** Long waits force engineers to run `git commit --no-verify`, letting flaws through.
- ❑ **Quota & Cost Burn:** 85%+ of commits are routine edits (docs, comments, formatting). Heavy frontier reasoning exhausts token allowances & Bobcoins instantly.
- ❑ **No deterministic secret gate:** Leaked AWS or GitHub keys wait for slow model inference instead of instant sub-millisecond halting.

```
$ git commit -m "docs: fix typo in README"
[System-2 Agent Thinking...] 24.8s elapsed... Quota: -1 Bobcoin
```

WITH S1GATE SYSTEM-1

DUAL-PROCESS FLOW

< 1.2s

Sub-Second Pre-Commit Verification Gate

- ❑ **Sub-millisecond Secret Detection:** Deterministic Shannon entropy filter halts high-entropy credentials in **0.05 milliseconds**.
- ❑ **Zero Quota Burn on Safe Edits:** Routine documentation and refactor diffs get instant PASS greenlight without invoking heavy reasoning models.
- ❑ **Developer Flow Preserved:** Engineers keep their rhythm while repositories stay strictly shielded from security vulnerabilities.

```
$ git commit -m "docs: fix typo in README"
[S1Gate] PASS (Score: 0) • Invariants clean • Time: 1.14s (0 Quota)
```

System-1 Intuition Meets System-2 Deliberation

Inspired by Daniel Kahneman's dual-process cognitive model: S1Gate separates sub-millisecond deterministic reflex from focused sub-second neural triage.

01 Shannon Scanner

High-entropy string detector and regex engine. Scans diff additions for AWS secret keys, GitHub tokens, Slack Webhooks, and private keys. Instant deterministic short-circuit.

< 0.05 ms

Fast Pre-Filter

02 AST Diff Parser

Parses unified diff syntax, strips lockfiles (package-lock, poetry.lock, uv.lock), and extracts semantic code chunks. Limits prompt footprint to under 8,000 characters.

< 0.02 ms

Zero Allocations

03 System-1 Triage

Evaluates code changes against security invariants using Google Gemini 3.1 Flash Lite (Keep-Alive pooled) or 100% offline Local LM Studio. Outputs structured Pydantic v2 JSON.

~1.2 s

Cloud or GPU

04 Policy Gate

Tri-state enforcement: **PASS** (Score 0-24, zero quota used), **WARN** (Score 25-69, contract alert), or **BLOCK** (Score 70-100, security flaw). Triggers Actor-Critic loop on BLOCK.

Tri-State

PASS / WARN / BLOCK

Actor–Critic Remediation with IBM Bob

When a vulnerability is caught, S1Gate doesn't just block — it empowers IBM Bob's Agent mode to diagnose, repair, and verify the patch automatically.

1 S1Gate Blocks Vulnerable Diff

Commit is halted. Generates structured Remediation Intent Protocol (RIP) task at `.bob/tasks/pending_remediation.json`.

2 IBM Bob Agent Reads Task Packet

Bob reads the vulnerability diagnosis, CWE classification, and targeted remediation hint. Inspects source via `slgate_inspect_file`.

3 Bob Synthesizes Surgical Fix (System–2)

Bob uses its deep deliberative reasoning to patch the vulnerability (e.g., parameterizing SQL, sanitizing paths, fixing race condition).

4 S1Gate Critic Verifies & Clears Commit

Bob invokes `slgate_verify_remediation` MCP tool. Risk score drops from 95 to 0. Commit is cleared and finalized!

`.bob/tasks/pending_remediation.json`

RIP TASK PACKET

```
{
  "task_id": "rip-2026-cve-891",
  "decision": "BLOCK",
  "original_risk_score": 95,
  "violated_invariants": ["has_ssrf", "flawed_ip_resolution"],
  "affected_file": "services/webhook_client.py",
  "remediation_hint": "Validate resolved IP against RFC1918 and link-local ranges before socket connection.",
  "critic_tool": "slgate_verify_remediation"
}
```

Status: Verified Fixed • Risk: 0/100 • Safe to Commit

Zero Escapes: 100% Catch Across 12 Production CVEs

Tested against realistic multi-language git diffs (Python, TypeScript, Go, Java) designed by an unbiased security auditor.

12 / 12

Flaws Blocked (100%)

0

Type II Escapes (Zero Leakage)

0.05 ms

Entropy Secret Detection

85 / 85

Automated Tests Passing

CVE / Flaw Category	Language	Vulnerability Description	S1Gate Decision	Detection Time
CWE-918 (SSRF)	Python	IPv4-mapped IPv6 internal cloud metadata bypass	BLOCKED (Score: 92)	1.21s
CWE-1321 (Prototype Pollution)	TypeScript	Recursive deep merge touching __proto__ object prototype	BLOCKED (Score: 94)	1.18s
CWE-639 (BOLA / IDOR)	Python/FastAPI	Multi-tenant resource retrieval missing org_id boundary	BLOCKED (Score: 90)	1.25s
CWE-362 (Race Condition / TOCTOU)	Go	Unsynchronized wallet balance debit without mutex lock	BLOCKED (Score: 88)	1.14s
CWE-502 (Insecure Deserialization)	Python	Unsafe PyYAML FullLoader cache deserialization leading to RCE	BLOCKED (Score: 96)	1.20s
CWE-798 (Buried Secrets)	Python / Env	High-entropy AWS Secret Key disguised in mock config	BLOCKED (Entropy)	0.04ms

Live Remote MCP Server & Ready for Judges

S1Gate is deployed in production behind Cloudflare reverse proxy with zero-install setup for IBM Bob IDE and Claude Code.

Production Endpoints

REMOTE MCP SSE ENDPOINT:

`https://mcp.cuang.dev/s1gate/sse`

HEALTH MONITORING:

`https://mcp.cuang.dev/s1gate/health`

JUDGE EVALUATION API KEY:

`s1gate-judge-secret-key-2026`

✂ **Sub-1.5s Response Time:** Streaming SSE + JSON Keep-Alive pooling with real-time latency reporting headers.

IBM Bob mcp.json Configuration

ZERO LOCAL SETUP

```
{
  "mcpServers": {
    "s1gate": {
      "url": "https://mcp.cuang.dev/s1gate/sse",
      "headers": {
        "Authorization": "Bearer s1gate-judge-secret-key-2026"
      }
    }
  }
}
```

Repository: [WilliamAxelC/IBM-hackathon](https://github.com/WilliamAxelC/IBM-hackathon)

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